

Python Dictionaries Complete Tutorial

Python Dictionaries – Complete Tutorial

This notebook covers **Python Dictionaries** and **all dictionary functions** with examples.

1. Introduction to Dictionaries

A dictionary stores data in **key-value pairs**.

```
student = {"name": "Ali", "age": 20, "cgpa": 3.6}
print(student)
```

```
{'name': 'Ali', 'age': 20, 'cgpa': 3.6}
```

2. Creating Dictionaries

```
#Simple Method
d1 = {"a": 1, "b": 2}

#Using Dict Function
d2 = dict(name="Sara", marks=85)

#Using Dict Function and List of Tuples
d3 = dict([("x", 10), ("y", 20)])

#Using Zip function with Dict Function to combine two lists into a dictionary
keys = ["name", "age", "city"]
values = ["Ali", 25, "Karachi"]
d4 = dict(zip(keys, values))

#Creating a Dictionary from a sequence of keys with the same default value.
keys = ["a", "b", "c"]
d5 = dict.fromkeys(keys, 0) # All keys get value 0

print(d1, d2, d3, d4, d5)
```

```
{'a': 1, 'b': 2} {'name': 'Sara', 'marks': 85} {'x': 10, 'y': 20} {'name': 'Ali', 'age': 25, 'city': 'Karachi'} {'a': 0, 'b': 0, 'c': 0}
```

3. Accessing Values

```
#Access by Key: Fastest, Returns Error if key doesn't exist
print(student["name"])

#If key doesn't exist, then a Message is printed, Rather than a key error.
print(student.get("department", "Not Assigned"))

#Access All Values as a object of Dictionary values, then we may cast it to a list
print(student.values())
```

```
Ali
Not Assigned
dict_values(['Ali', 20, 3.6])
```

4. Adding and Updating Values

```
#Direct Assignment
student["department"] = "AI"

#Update Method: Add multiple key-value pairs at once. Updates existing keys if
present.
student.update({"department": "AI", "semester": 5})
print(student)

#setdefault() Method: Add key with a default value only if it doesn't exist.
student.setdefault("section", "A") # Adds 'section': 'A'
student.setdefault("age", 21)      # Won't overwrite existing key
print(student)

#Update Using Key
student["age"] = 21
print(student)

#Merging Two Dictionaries
additional_info = {"hobby": "reading", "city": "Karachi"}
student = student | additional_info
print(student)
```

```
{'name': 'Ali', 'age': 20, 'cgpa': 3.6, 'department': 'AI', 'semester': 5}
{'name': 'Ali', 'age': 20, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A'}
{'name': 'Ali', 'age': 21, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A'}
{'name': 'Ali', 'age': 21, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A', 'hobby': 'reading', 'city': 'Karachi'}
```

5. Dictionary Functions / Methods

```
print(len(student))          # len()
print(student.keys())        # keys()
print(student.values())      # values()
print(student.items())       # items()
```

```
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dict_keys(['name', 'age', 'cgpa', 'department', 'semester', 'section', 'hobby', 'city'])
dict_values(['Ali', 21, 3.6, 'AI', 5, 'A', 'reading', 'Karachi'])
dict_items([('name', 'Ali'), ('age', 21), ('cgpa', 3.6), ('department', 'AI'), ('semester', 5), ('section', 'A'), ('hobby', 'reading'), ('city', 'Karachi')])
```

```
#Pop Removes and Returns the value
removed=student.pop("city")
print(removed)
print(student)
```

```
Karachi
{'name': 'Ali', 'age': 21, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A', 'hobby': 'reading'}
```

```
#Removes Last Inserted Value
item = student.popitem()
print(item)
print(student)
```

```
#Returns a Copy
copy_student = student.copy()
copy_student["age"] = 25
print(student)
print(copy_student)
```

```
{'name': 'Ali', 'age': 21, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A', 'hobby': 'reading'}
{'name': 'Ali', 'age': 25, 'cgpa': 3.6, 'department': 'AI', 'semester': 5, 'section': 'A', 'hobby': 'reading'}
```

```
#Populate a Dictionary via keys
keys = ["id", "name", "salary"]
emp = dict.fromkeys(keys, "None")
print(emp)
```

```
{'id': 'None', 'name': 'None', 'salary': 'None'}
```

6. Looping Through Dictionary

```
for k, v in student.items():
    print(k, ":", v)
```

```
name : Ali
age : 21
cgpa : 3.6
department : AI
semester : 5
section : A
hobby : reading
```

7. Nested Dictionaries

```
university = {
    "AI": {"students": 120, "labs": 3},
    "CS": {"students": 200, "labs": 4}
}
print(university["AI"]["students"])
```

```
120
```

8. Dictionary Comprehension

```
squares = {x: x*x for x in range(1, 6)}
print(squares)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

9. Practical Example – Frequency Counter

```
sentence = "python python ai ml ai python"
freq = {}

for word in sentence.split():
    freq[word] = freq.get(word, 0) + 1

print(freq)
```

10. Clear Dictionary

```
student.clear()
print(student)
```

 **End of Dictionary Tutorial Notebook**